Lecture 8: Identifiability and Fisher information

COMP 521 • Probability, Optimization, and Scientific Discovery • Sections 4.1–4.4

This handout is one part of the full COMP 521 notes. Section, equation, figure and listing numbers match the full notes, and references to anything outside this handout point there. The notation table is in "How to use these notes" (Lecture 1).

4 Identifiability and scientific discovery

4.1 Sensitivity is not identifiability

A parameter is *sensitive* if changing it changes a prediction. It is *identifiable* if the data can tell its values apart. These are different properties.

Worked example 4.1 (A perfect degeneracy)**.** Let a normalized observable depend on two TOY mixing coefficients through $y_1 = \theta_1 + \theta_2$. Observing $y_1 = 3$ is satisfied by the whole line $\theta_2 = 3 - \theta_1$. The observable is sensitive to each parameter, yet neither can be determined. Adding $y_2 = \theta_1 - \theta_2 = 1$ singles out $(\theta_1, \theta_2) = (2, 1)$.

Two levels of the idea should be kept apart:

Definition 4.2 (Structural and practical identifiability)**.** A model is *structurally* (globally) identifiable if $p(\mathbf{d} \mid \boldsymbol{\theta}) = p(\mathbf{d} \mid \boldsymbol{\theta}')$ for all $\mathbf{d}$ implies $\boldsymbol{\theta} = \boldsymbol{\theta}'$: with perfect, noise-free data of the planned kind, the parameters could be recovered. It is *locally* structurally identifiable at $\boldsymbol{\theta}$ if this holds in a neighbourhood. A parameter is *practically* identifiable if, with the finite, noisy data actually available, its uncertainty is small enough to answer the scientific question.

$y_1 = \theta_1 + \theta_2$ alone is structurally unidentifiable: no amount of data of that kind helps. Two nearly parallel observables, for instance $y_1 = \theta_1 + \theta_2$ and $y_2 = \theta_1 + 1.05\,\theta_2$, are structurally identifiable but may be practically unidentifiable: in principle the data separate the parameters, in practice the errors would have to be tiny. The remedies differ. Structural problems need a different *kind* of observable; practical problems may be solved by more or better data of the same kind.

For stellar evolution, luminosity alone can trade mass against metallicity, age and mixing; abundances or asteroseismic frequencies may constrain other combinations. Whether they do depends on the Jacobians and the realistic errors, which is a calculation, not an intuition.

4.2 Fisher information

Definition 4.3 (Fisher information)**.** The *score* is the gradient of the log-likelihood, $\nabla_{\boldsymbol{\theta}} \ell$. The Fisher information matrix is its covariance under the sampling distribution of the data,

$$\mathcal{I}(\boldsymbol{\theta}) = \mathbb{E}_{\mathbf{d}|\boldsymbol{\theta}}\big[\nabla \ell \, \nabla \ell^{\mathsf{T}}\big] = -\mathbb{E}_{\mathbf{d}|\boldsymbol{\theta}}\big[\nabla^2 \ell\big], \tag{32}$$

the second equality holding under standard regularity conditions (support independent of $\boldsymbol{\theta}$, differentiation under the integral).

For Gaussian data $\mathbf{d} \sim \mathcal{N}(\mathcal{G}(\boldsymbol{\theta}), \boldsymbol{\Sigma})$ with $\boldsymbol{\Sigma}$ fixed,

$$\mathcal{I}(\boldsymbol{\theta}) = \mathbf{J}(\boldsymbol{\theta})^{\mathsf{T}} \boldsymbol{\Sigma}^{-1} \mathbf{J}(\boldsymbol{\theta}), \qquad \mathbf{J} = \partial \mathcal{G} / \partial \boldsymbol{\theta}. \tag{33}$$

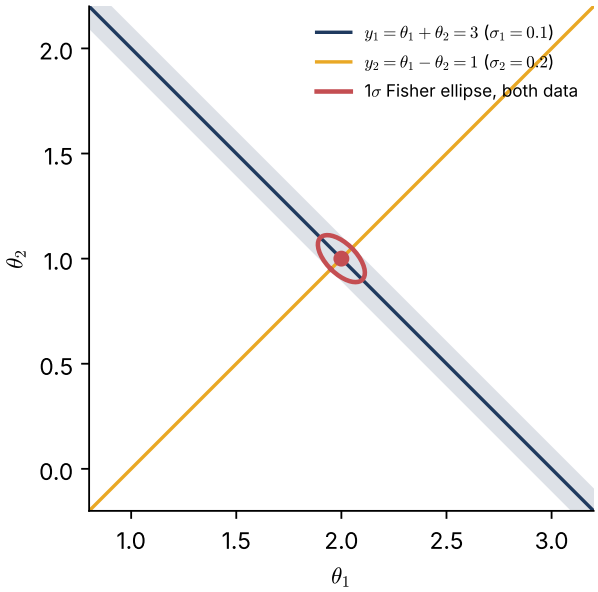


Figure 7: Each observable alone constrains a band of parameter space (shaded for y_1). Complementary observables intersect in a closed ellipse, the 1σ contour of the inverse Fisher matrix of Section 4.2.

If the covariance depends on parameters, add $\frac{1}{2}\,\mathrm{tr}(\boldsymbol{\Sigma}^{-1}\partial_i\boldsymbol{\Sigma}\,\boldsymbol{\Sigma}^{-1}\partial_j\boldsymbol{\Sigma})$ to entry (i,j). Fisher matrices from *independent* data sets add, which is what makes them a planning tool.

Worked example 4.4 (The degeneracy, in matrices)**.** For $y_1 = \theta_1 + \theta_2$ with $\sigma_1 = 0.1$: $\mathbf{J}_1 = (1\ \ 1)$ and $\mathcal{I}_1 = 100\left(\begin{smallmatrix}1&1\\1&1\end{smallmatrix}\right)$, with eigenvalues 200 and 0. The zero eigenvalue's eigenvector $\propto (1,-1)^{\mathsf{T}}$ is the direction the data cannot see. Adding $y_2 = \theta_1 - \theta_2$ with $\sigma_2 = 0.2$ contributes $25\left(\begin{smallmatrix}1&-1\\-1&1\end{smallmatrix}\right)$, so $\mathcal{I} = \left(\begin{smallmatrix}125&75\\75&125\end{smallmatrix}\right)$ with eigenvalues 200 and 50: both directions are now informed.

A rank-deficient Fisher matrix signals local structural non-identifiability [1]; a nearly singular one signals that some combination will be poorly constrained.

Result 4.5 (Cramér–Rao bound)**.** Under regularity conditions, any *unbiased* estimator $\hat{\boldsymbol{\theta}}$ satisfies $\mathrm{Cov}(\hat{\boldsymbol{\theta}}) \succeq \mathcal{I}(\boldsymbol{\theta})^{-1}$ (the difference is positive semi-definite).

Three cautions. The bound concerns unbiased estimators, and biased ones (including every Bayesian posterior mean with an informative prior) can beat it. It is local, so it misses multimodality, boundaries and heavy tails. And when $\mathcal{I}$ is singular, $\mathcal{I}^{-1}$ does not exist, and no unbiased estimator of the unconstrained combination exists at all. Then report the constrained combinations, or add prior information, which in the Gaussian case gives the bound $(\mathcal{I} + \boldsymbol{\Sigma}_0^{-1})^{-1}$.

4.2.1 Fisher eigenvectors depend on your units

A tempting reading of $\mathcal{I}$ is "its stiffest eigenvector is the parameter combination the data measure best". That reading depends on the units of the parameters. If $\boldsymbol{\theta}' = \mathbf{S}^{-1}\boldsymbol{\theta}$ for a diagonal rescaling $\mathbf{S}$, then $\mathcal{I}' = \mathbf{S}\mathcal{I}\mathbf{S}$, whose eigenvectors generally point elsewhere: an eigen-decomposition asks a question about Euclidean distance in parameter space, and Euclidean distance between α_{MLT} and f_{ov} has no physical meaning.

The fix is to measure information *relative to what you already knew*. Solve the generalized eigenproblem

$$\mathcal{I}\,\mathbf{v} = \lambda\,\boldsymbol{\Sigma}_0^{-1}\mathbf{v} \quad\Longleftrightarrow\quad (\mathbf{L}^\mathsf{T}\mathcal{I}\mathbf{L})\mathbf{u} = \lambda\mathbf{u}, \qquad \boldsymbol{\Sigma}_0 = \mathbf{L}\mathbf{L}^\mathsf{T}. \tag{34}$$

The eigenvalues λ_i are invariant under any linear reparameterization (the prior widths rescale with the parameters). They have a direct reading: $\lambda_i > 1$ means the data dominate the prior along that direction; $\lambda_i \ll 1$ means the data add almost nothing there.

`ex05_fisher_units.py -- Fisher eigenvectors before and after changing the units of` f_{ov}

```python
"""Fisher eigenvectors depend on parameter units; prior-whitened ones do not.

Toy linearized model with two hypothetical physics parameters:
  theta1 = mixing-length parameter alpha_MLT   (order 1)
  theta2 = overshoot parameter f_ov            (order 0.01)
Two observables with an assumed (toy) Jacobian and independent Gaussian errors.
"""
import jax
import jax.numpy as jnp
from jax.scipy.linalg import eigh

jax.config.update("jax_enable_x64", True)

J = jnp.array([[0.30, 12.0],    # d(obs1)/d(alpha), d(obs1)/d(f_ov)  [toy numbers]
               [0.10, 10.0]])   # d(obs2)/d(alpha), d(obs2)/d(f_ov)
Sigma_inv = jnp.diag(1.0 / jnp.array([0.02, 0.05]) ** 2)
prior_sd = jnp.array([0.2, 0.01])  # prior widths in the ORIGINAL units


def report(label, F, prior_sd):
    w, V = jnp.linalg.eigh(F)
    print(f"{label}: Fisher eigenvalues {w}, stiffest direction {V[:, -1]}")
    # prior-whitened problem: F v = lambda Sigma0^-1 v  <=>  eig of D F D, D = diag(prior_sd)
    D = jnp.diag(prior_sd)
    lam, _ = eigh(D @ F @ D)
    print(f"{'':>{len(label)}}  prior-whitened eigenvalues {lam}  (>1: data beat the prior)")


F = J.T @ Sigma_inv @ J
report("f_ov in raw units      ", F, prior_sd)

# Re-express f_ov in units of 0.01:  theta2' = theta2 / 0.01, so d/dtheta2' = 0.01 d/dtheta2
S = jnp.diag(jnp.array([1.0, 0.01]))
F_scaled = S @ F @ S
report("f_ov in units of 0.01  ", F_scaled, prior_sd / jnp.array([1.0, 0.01]))
```

Output (executed for these notes):

```
f_ov in raw units      : Fisher eigenvalues [8.09552915e+00 4.00220904e+05], stiffest direction [0.02349399
     0.99972398]
                         prior-whitened eigenvalues [ 0.26505809 48.89494191]  (>1: data beat the prior)
f_ov in units of 0.01  : Fisher eigenvalues [  1.20990284 267.79009716], stiffest direction [-0.92438626
     -0.38145779]
                         prior-whitened eigenvalues [ 0.26505809 48.89494191]  (>1: data beat the prior)
```

Changing f_{ov}'s units rotates the stiffest eigenvector from almost pure f_{ov} to mostly α_{MLT}, and changes the eigenvalues by three orders of magnitude, yet nothing about the experiment changed. The prior-whitened eigenvalues (0.27 and 48.9) are identical: along one combination the data carry

about 49 times the prior's precision (so the posterior standard deviation there is $\sqrt{1+48.9} \approx 7$ times smaller than the prior's), and the other is barely informed.

4.3 Beyond local: profile likelihoods

Fisher analysis linearizes. To test practical identifiability without linearizing, compute the profile likelihood $\ell_{\text{prof}}(\theta_j) = \max_{\boldsymbol{\theta}_{-j}} \ell(\boldsymbol{\theta})$ on a grid of θ_j by repeated optimization [2]. A parameter is practically identifiable at 95% confidence if $-2[\ell_{\text{prof}}(\theta_j) - \ell_{\max}]$ rises above 3.84, the 95th percentile of a χ^2 distribution with one degree of freedom, on both sides of the optimum. A profile that flattens on one side (for example toward large f_{ov}) reveals a one-sided constraint that a Fisher ellipse would draw as symmetric.

4.4 Model discrepancy and its own identifiability problem

Every stellar model is wrong somewhere. Kennedy and O'Hagan [3] made this explicit by writing observations as

$$\mathbf{d} = \mathcal{G}(\boldsymbol{\theta}) + \boldsymbol{\delta} + \boldsymbol{\epsilon}, \tag{35}$$

with $\boldsymbol{\delta}$ an unknown, structured discrepancy (often a Gaussian process over the conditions of the observation) and $\boldsymbol{\epsilon}$ the measurement noise. Including $\boldsymbol{\delta}$ prevents a fit from forcing physical parameters to unphysical values to compensate for missing physics.

Discrepancy and parameters are confounded

A flexible $\boldsymbol{\delta}$ can mimic the effect of changing $\boldsymbol{\theta}$, so $\boldsymbol{\theta}$ and $\boldsymbol{\delta}$ are generally not jointly identifiable: the posterior on $\boldsymbol{\theta}$ becomes broad or depends strongly on the prior for $\boldsymbol{\delta}$ [4]. The remedy is physical knowledge: informative priors on the size and shape of the discrepancy (for example, "missing transport physics can change T_{eff} on the giant branch by up to $\sim 100\,\text{K}$ but not the main-sequence luminosity"), and data whose relation to $\boldsymbol{\theta}$ the discrepancy cannot mimic.

This is why scientific claims should be stated conditionally: "within this family of stellar models, with these conventional uncertainties, the data constrain this combination of parameters to this range". Repeating the analysis with alternative prescriptions for convection, mass loss and composition is how one learns whether a conclusion survives the model's known weaknesses.

Exercises for this lecture

Exercises 14, 15, 17, numbered as in Appendix A of the full notes. Worked solutions are in Appendix B; attempt each one before reading its solution.

14. [P] Nearly degenerate is not degenerate. Observables $y_1 = \theta_1 + \theta_2$ and $y_2 = \theta_1 + 1.05\,\theta_2$, each with $\sigma = 0.1$. Compute $\mathcal{I}$, its eigenvalues and condition number, and the Cramér–Rao standard deviations of θ_1 and θ_2. Is the model structurally identifiable? Practically?

15. [P] Fisher for counts. You count $N \sim \text{Poisson}(\theta\, T)$ photons in exposure time T. Find $\mathcal{I}(\theta)$ and the Cramér–Rao bound. Does the MLE attain it? How does the bound change if T doubles?

17. [R] Reading a Fisher matrix honestly. A student computes the Fisher matrix for $(\alpha_{\text{MLT}}, f_{\text{ov}})$ from asteroseismic data and writes "the stiffest eigenvector is almost pure f_{ov}, so these data

mainly constrain overshooting". What question would you ask first? What calculation would settle it?

References

[1] T. J. Rothenberg. Identification in parametric models. *Econometrica*, 39(3):577–591, 1971.

[2] A. Raue, C. Kreutz, T. Maiwald, J. Bachmann, M. Schilling, U. Klingmüller, and J. Timmer. Structural and practical identifiability analysis of partially observed dynamical models by exploiting the profile likelihood. *Bioinformatics*, 25(15):1923–1929, 2009.

[3] M. C. Kennedy and A. O'Hagan. Bayesian calibration of computer models. *Journal of the Royal Statistical Society: Series B*, 63(3):425–464, 2001.

[4] J. Brynjarsdóttir and A. O'Hagan. Learning about physical parameters: the importance of model discrepancy. *Inverse Problems*, 30:114007, 2014.